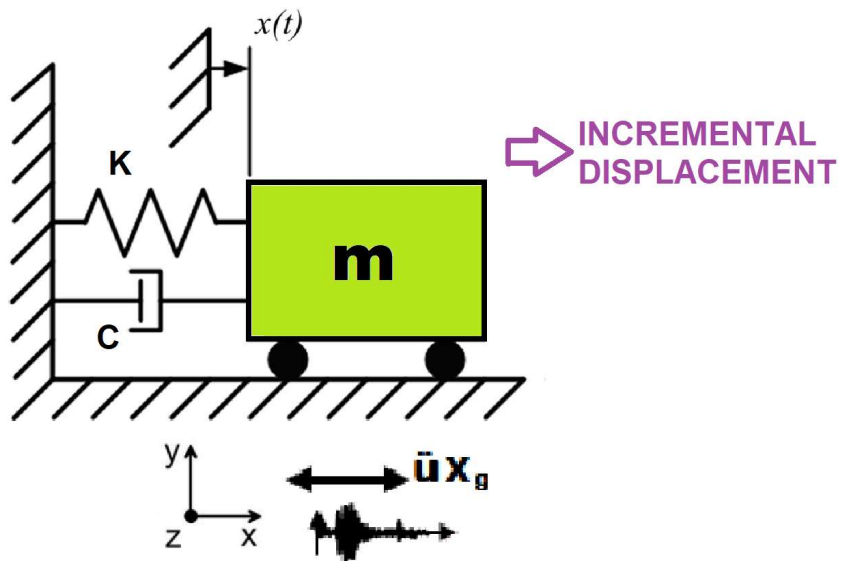


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**DETERMINATION OF OPTIMUM
STRUCTURAL ULTIMATE
DISPLACEMENT FROM ENERGY
DISSIPATION CAPACITY INDEX USING
FINITE-DIFFERENCE NEWTON
ITERATION AND OPENSEES**

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

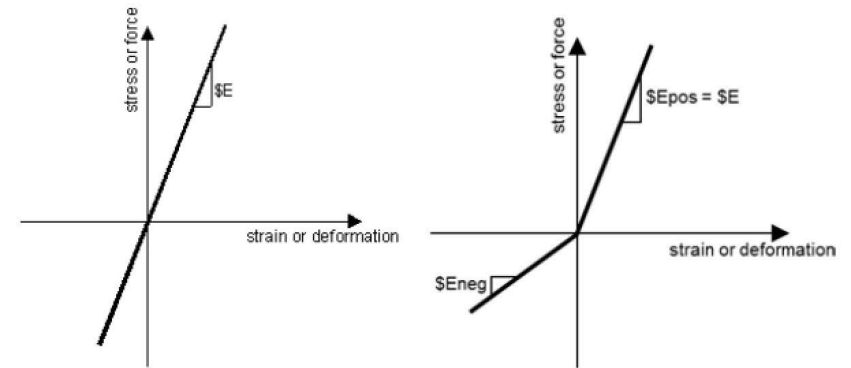
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

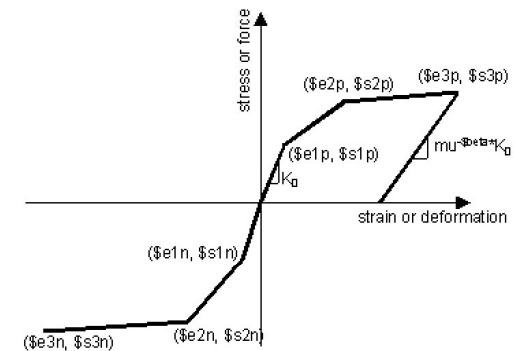
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

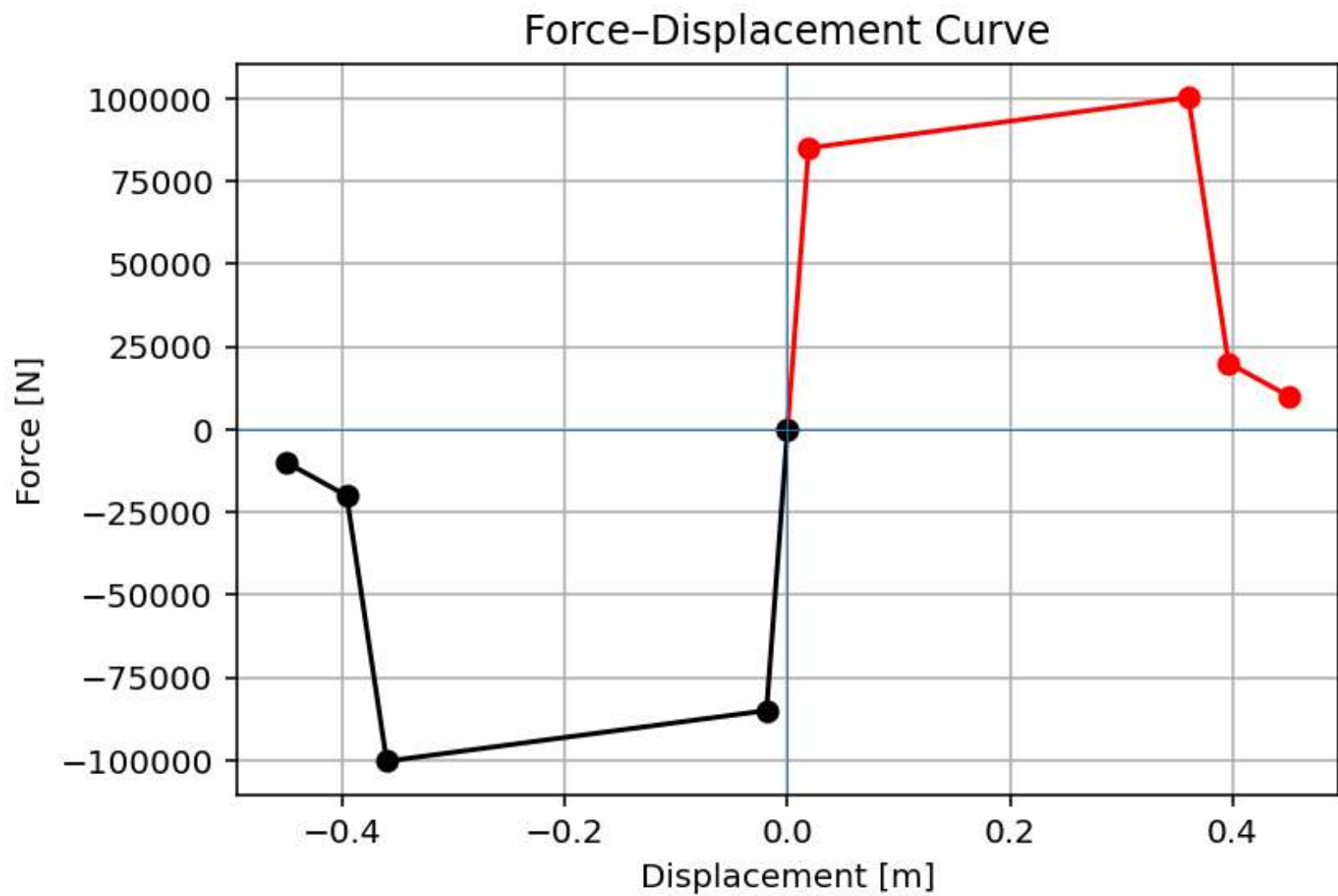
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL



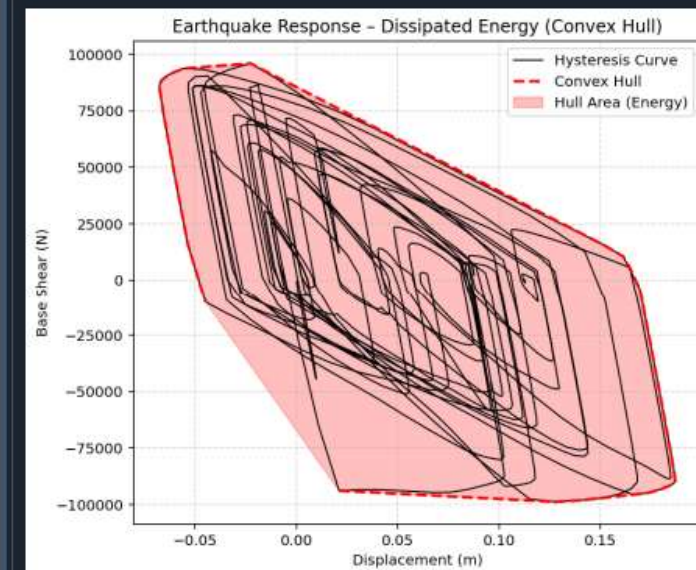
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...IMIZATION_DECI_SDOF\DSU_OPTIMIZATION_DECI_SDOF.py

DSU_OPTIMIZATION_DECI_SDOF.py X DISSIPATED_ENERGY_WITH_PLOT_FUN.py X

```

1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # DETERMINATION OF OPTIMUM STRUCTURAL ULTIMATE DISPLACEMENT FROM ENERGY DISSIPATION CAPACITY
4 # FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES
5 #
6 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM STRUCTURE : AN OPEN
7 # FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY
8 #
9 #
10 #  $P(t) = P_0 \sin(\omega t)$ 
11 #  $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$ 
12 #
13 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION
14 # OF ENERGY DISSIPATION CAPACITY INDEX
15 #
16 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
17 # EMAIL: salar.d.ghashghaei@gmail.com
18 #####
19 """
20 Nonlinear Seismic Performance Assessment of a SDOF:
21 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and E
22
23 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a SDOF
24 for performance-based earthquake engineering. The 2D model incorporates both material nonl
25 (elastic-perfectly plastic or hysteretic steel with strain hardening)
26 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
27 and large displacements-essential for collapse assessment.
28
29 Eight analysis protocols are implemented:
30 (1) [PERIOD] : Structural Period
31 (2) [STATIC] : Gravity load analysis establishing dead load state
32 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
33 and plastic mechanism identification
34 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
pinching behavior, and energy dissipation degradation

```



OPTIMIZATION PROBLEM FOR ENERGY DISSIPATION CAPACITY INDEX (EDCI)

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...IMIZATION_DECI_SDOF\DSU_OPTIMIZATION_DECI_SDOF.py

DSU_OPTIMIZATION_DECI_SDOF.py X DISSIPATED_ENERGY_WITH_PLOT_FUN.py X

```
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```

Console 1/A X

```
+-----+
| lambda | omega | period | frequency |
+-----+
| 9.000e+02 | 30.0000 | 0.2094 | 4.7746 |
+-----+

Time: 20.0000, Displacement: 0.1123 mm, Reaction: -20.00 N

SEISMIC ANALYSIS DONE.

Exact Damping Ratio: 5.97783688e-03
Xmax: 0.6154450173008287
Dissipated Energy from Earthquake= 31985.82 N·m
Dissipated Energy from Cyclic Displacement= 160197.34 N·m

DISSIPATED ENERGY CAPACITY INDEX: 19.967 [%]

ZONE 2: MINOR DAMAGE

Fmax: -0.03348853108128935
DF: -33.774669186451334
DX: -1.2037852705051022e-07
IT: 7 - RESIDUAL: 1.2037852705051022e-07 X: 0.6144450173008287

Optimum X (ULTIMATE DISPLACEMENT): 0.6144
Iteration Counts: 7
Convergence Residual: 1.2037852705e-07

Total time (s): 66.0312
```

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 146, Col 20 UTF-8 CRLF RW Mem 54%

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DISSIPATED ENERGY CAPACITY INDEX: 19.967 [%]

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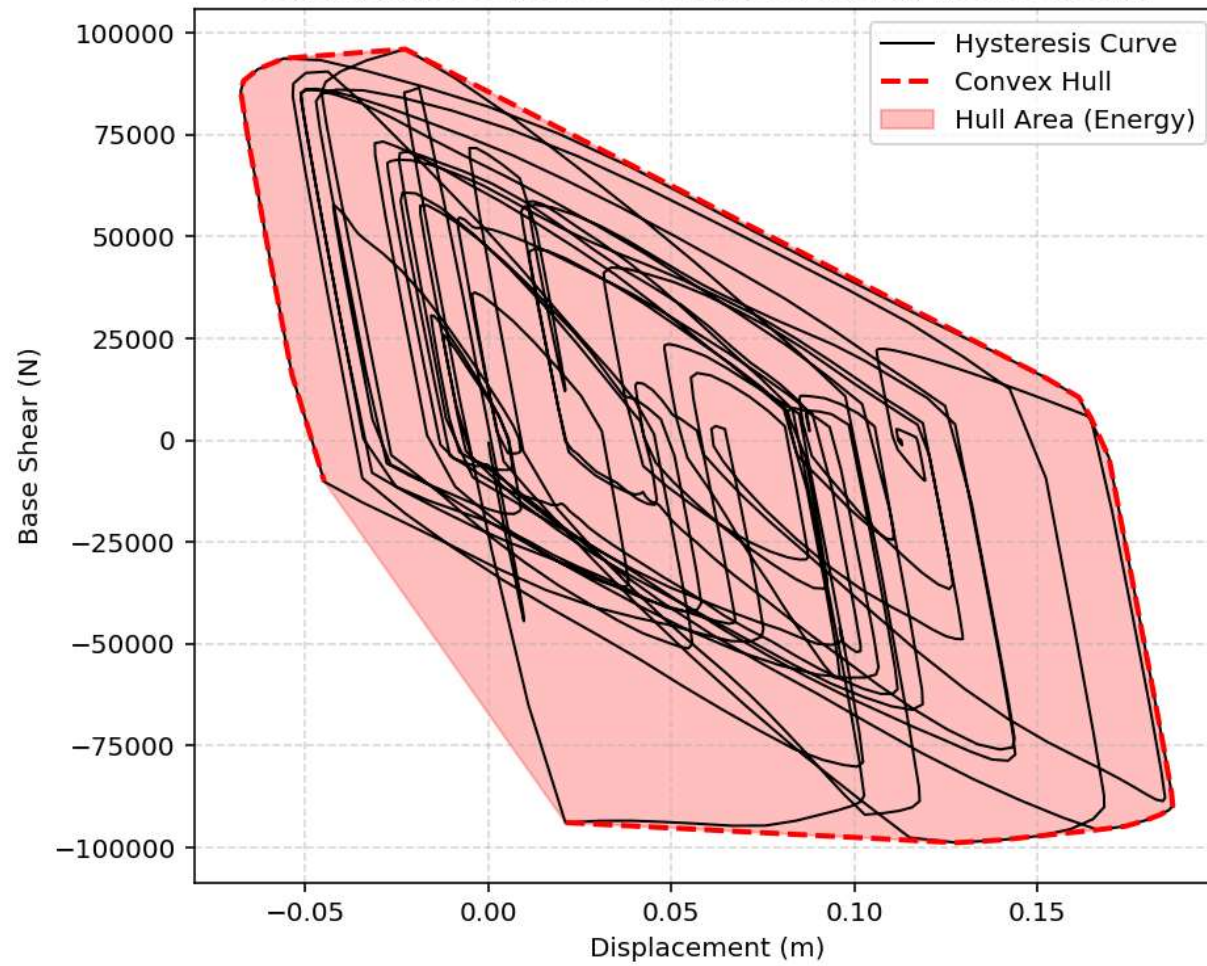
Optimum X (ULTIAMTE DISPLACEMENT): 0.6144

Iteration Counts: 7

Convergence Residual: 1.2037852705e-07

Total time (s): 66.0312

Earthquake Response - Dissipated Energy (Convex Hull)



Cyclic Loading - Dissipated Energy (Convex Hull)

